

Section 2.1 – Principles of Network Applications

❖ Client-server architecture vs. Peer-to-peer architecture

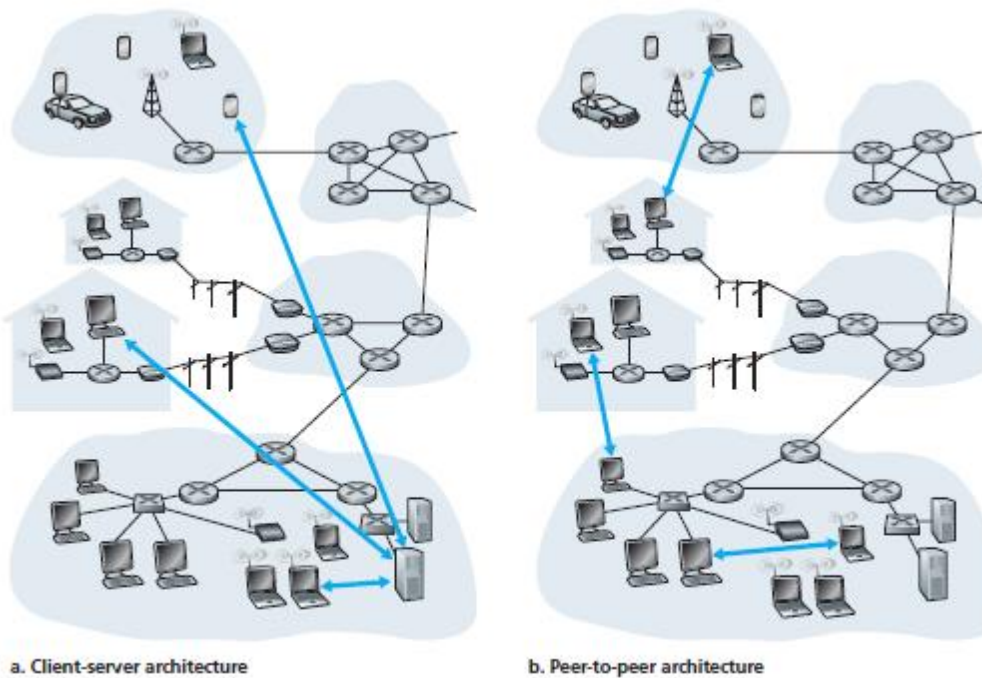


Figure 2.2 ♦ (a) Client-server architecture; (b) P2P architecture

❖ Process Communication – Client Process and server Process

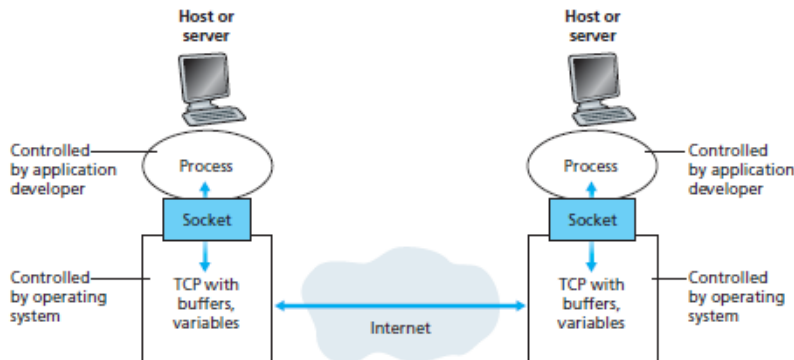
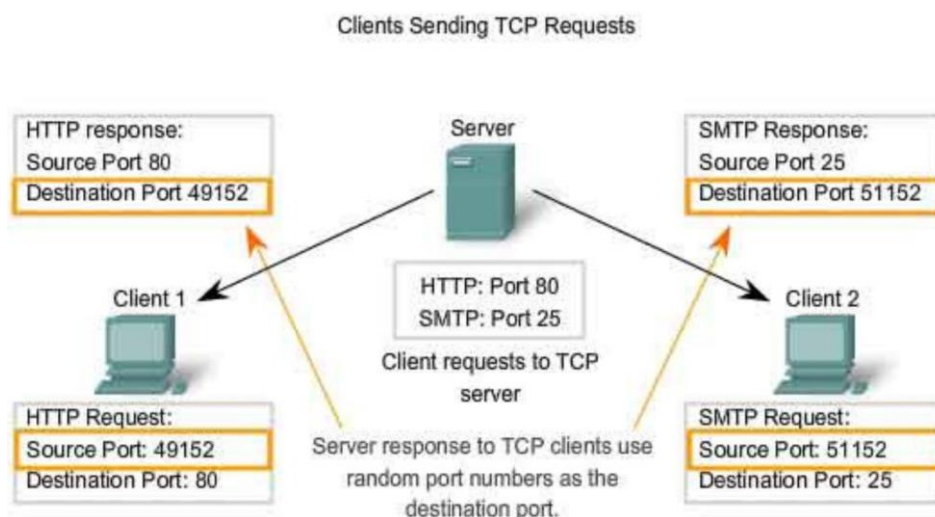
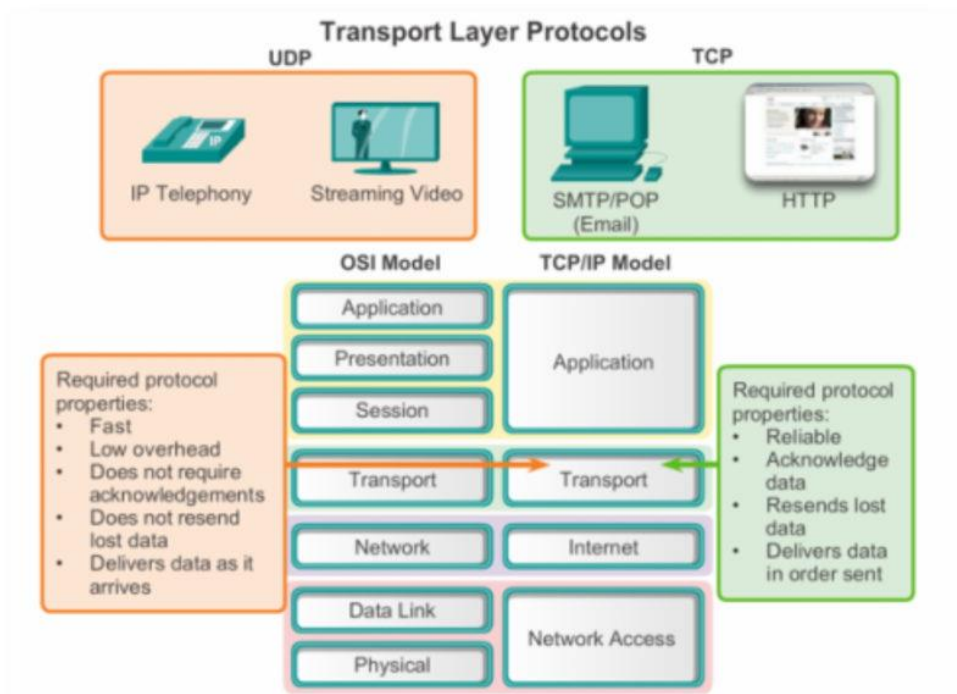


Figure 2.3 ♦ Application processes, sockets, and underlying transport protocol



❖ Transport services provided by the Internet – TCP and UDP



❖ What an Application layer protocol defines? (4 things)

Knowledge Test # 1

Q.1 Write down **3 (Three)** distinct differences between **Client-server** and **Peer-to-peer** architecture.

Client-Server	Peer-to-peer

Q.2 What **information** is used by a process running on one host to identify a process running on another host?

Q.3 Suppose you wanted to do a transaction from a remote client to a server **as fast as possible**. Would you use **UDP** or **TCP**? **Why**?

Q.4 List the **four broad classes of services** that a transport protocol can provide. For each of the service classes, indicate if either UDP or TCP (or both) provides such a service. (Hints: **Data loss**, **Timing**, **Bandwidth** and **Security**)

Q.5 What are the **four things** defined in an **application layer protocol**?

Section 2.2 – The Web and HTTP



❖ Web page consists of Objects

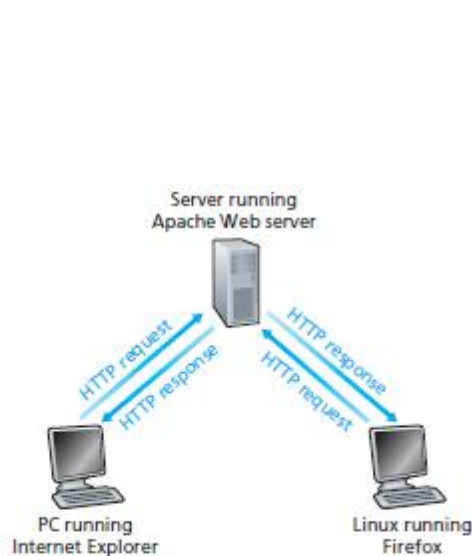


Figure 2.6 ♦ HTTP request-response behavior

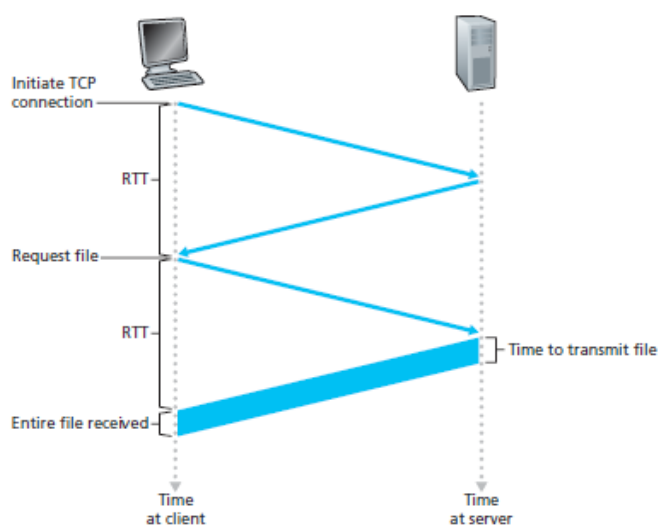


Figure 2.7 ♦ Back-of-the-envelope calculation for the time needed to request and receive an HTML file

❖ HTTP Connections – Non-persistent and Persistent (with/without pipeline)

❖ HTTP message types – HTTP request and HTTP response

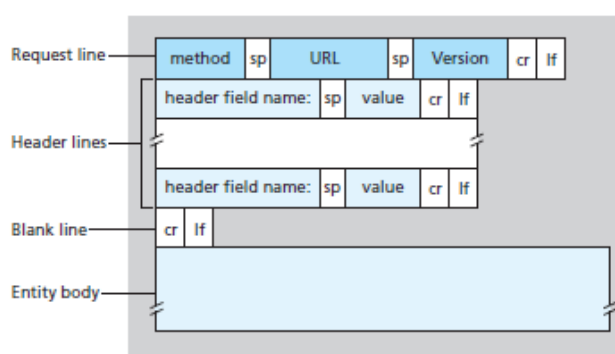


Figure 2.8 ♦ General format of an HTTP request message

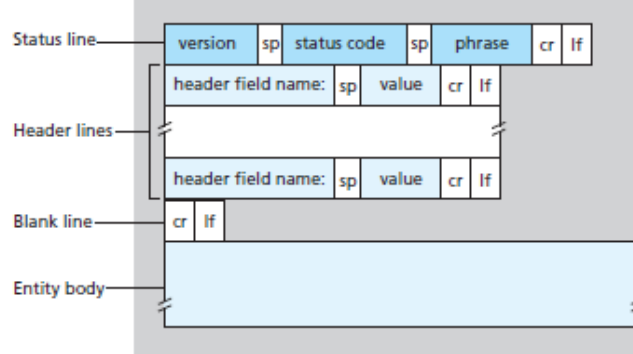


Figure 2.9 ♦ General format of an HTTP response message

Question

Assume that you have a base HTML file with 30 embedded images. How many RTTs are required to retrieve the base file and images under the following conditions?

- (i) Non-persistent connection without parallel connection
 - (ii) Non-persistent connection with 10 parallel connections
 - (iii) Persistent connection without pipelining
 - (iv) Persistent connection with pipelining
-

Key Definitions

RTT (Round Trip Time): Time for a packet to travel from client to server and back.

FTT (File Transmission Time): Time to push all bits of a file onto the link. $FTT = \text{File Size} / \text{Bandwidth}$.

Total objects: 1 base HTML file + 30 images = 31 objects

Per object cost (non-persistent):

$$1 \text{ RTT (TCP handshake)} + 1 \text{ RTT (HTTP request/response)} + \text{FTT} = 2 \text{ RTT} + \text{FTT}$$

Per object cost (persistent):

$$1 \text{ RTT (HTTP only, TCP shared)} + \text{FTT}$$

Solution

(i) Non-persistent connection, no parallel connections

Each object requires a new TCP connection:

$$1 \text{ RTT (TCP handshake)} + 1 \text{ RTT (HTTP)} + \text{FTT}$$

31 objects are fetched sequentially.

$$\text{Total} = 31 \times (2 \text{ RTT} + \text{FTT})$$

$$= 62 \text{ RTT} + 31 \text{ FTT}$$

(ii) Non-persistent connection, 10 parallel connections

Base HTML is fetched first (serial): $2 \text{ RTT} + \text{FTT}$

30 images fetched in rounds of 10 parallel connections:

- Round 1: 10 images $\Rightarrow 2 \text{ RTT} + \text{FTT}$

- Round 2: 10 images $\Rightarrow 2 \text{ RTT} + \text{FTT}$

- Round 3: 10 images $\Rightarrow 2 \text{ RTT} + \text{FTT}$

$$3 \text{ rounds} \times (2 \text{ RTT} + \text{FTT}) = 6 \text{ RTT} + 3 \text{ FTT}$$

$$\text{Total} = (2 \text{ RTT} + \text{FTT}) + 3 \times (2 \text{ RTT} + \text{FTT})$$

$$= 8 \text{ RTT} + 4 \text{ FTT}$$

(iii) Persistent connection, no pipelining

One TCP handshake shared for all objects: 1 RTT

Each object then fetched one at a time: 1 RTT + FTT per object

$$31 \times (1 \text{ RTT} + \text{FTT}) = 31 \text{ RTT} + 31 \text{ FTT}$$

$$\text{Total} = 1 \text{ RTT} + 31 \times (1 \text{ RTT} + \text{FTT})$$

$$= 32 \text{ RTT} + 31 \text{ FTT}$$

(iv) Persistent connection, with pipelining

1 RTT for TCP handshake

1 RTT + FTT for the base HTML file

All 30 image requests sent together in one pipeline burst: 1 RTT + 30 FTT

$$\text{Total} = 1 \text{ RTT} + (1 \text{ RTT} + \text{FTT}) + (1 \text{ RTT} + 30 \text{ FTT})$$

$$= 3 \text{ RTT} + 31 \text{ FTT}$$

Summary

Case	RTT	FTT
(i) Non-persistent, no parallel	62 RTT	+ 31 FTT
(ii) Non-persistent, 10 parallel	8 RTT	+ 4 FTT
(iii) Persistent, no pipelining	32 RTT	+ 31 FTT
(iv) Persistent, with pipelining	3 RTT	+ 31 FTT

Knowledge Test # 2

Q.1 True or false?

- a. A user requests a Web page that consists of some text and **three images**. For this page, the client will send **one request message** and receive **four response messages**.
- b. **Two distinct Web pages** (for example, `www.mit.edu/research.html` and `www.mit.edu/students.html`) can be sent over the **same persistent connection**.
- c. With **non-persistent connections** between browser and origin server, it is possible for a **single TCP segment** to carry **two distinct HTTP request messages**.
- d. The **Date:** header in the **HTTP response message** indicates when the object in the response was last modified.
- e. **HTTP response messages** never have an empty message body.

Q.2 What are the **message types** in HTTP? Describe **different fields** of the messages.

Q.3 The HTTP GET message

Consider the figure below, where a client is sending an **HTTP GET message** to a web server, `gaia.cs.umass.edu`.



Suppose the client-to-server **HTTP GET message** is the following:

```
GET /kurose_ross/interactive/quotation4.htm HTTP/1.1
Host: gaia.cs.umass.edu
Accept: text/plain, text/html, text/xml, image/gif, image/jpeg,
audio/basic, audio/mp4, video/wmv, video/mp4, application/*, */*
Accept-Language: en-us, en-gb;q=0.6, en;q=0.9, fr, fr-ch, zh, cs
If-Modified-Since: Fri, 08 Feb 2013 12:09:35 EST
User Agent: Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/535.11 (KHTML,
like Gecko) Chrome/17.0.963.56 Safari/535.11
```

Answer the following questions:

- ✓ What is the **name of the file** that is being retrieved in this **GET message**?
- ✓ What **version of HTTP** is the **client** running?
- ✓ What **formats of text, images, audio, and video** does the client browser prefer to receive?
- ✓ What **languages** is the browser indicating that it is willing to accept?
- ✓ Does the client already have a (possibly out-of-date) copy of the requested file? **Explain**.
- ✓ What is the **type of client browser** and the **client's operating system**?

Q.4 The HTTP REPLY message

Describe the following HTTP reply message:

```
HTTP/1.1 200 OK
Date: Sat, 8 Oct 2011 15:39:29
Server: Apache/1.3.0 (Unix)
Last-Modified: Wed, 7 Sep 2011 09:23:24
Content-Type: image/gif

(data data data data data ...)
```

Q.5 In this problem, you will compare **persistent** vs. **non-persistent HTTP**. Suppose your browser downloads a webpage. The **webpage html** object is **7 Kbits** in length and additionally contains **5 embedded images**, each **2.5 Kbits** in length. The webpage and the 5 images are all stored on the **same server**. We will abstract the **network path** between your browser and the web server as a **50 Mbps link**. Assume your browser has a **100 ms** RTT to the server, which includes propagation, processing, and queuing delays. Assume also that the **time to transmit a GET message is zero**. You should, however, account for the **time needed to setup up a TCP connection** (i.e., 1 RTT) and the **time to transmit** the html file and the embedded images.

- a) Assume **one non-persistent HTTP connection**. What is the **response time**? I.e., the time from when the user requests the webpage to the time when the webpage and its embedded images are displayed? Make sure you describe the various components that contribute to the response time.
- b) Assume **non-persistent HTTP** and that the browser can open **two parallel TCP connections** to the server. What is the **response time**?
- c) Assume **one persistent HTTP connection with no pipelining**. What is the **response time**?
- d) Assume **one persistent HTTP connection with pipelining**. What is the **response time**?
- e) Are **non-persistent HTTP connections** ever preferable to **persistent HTTP connections**?

Section 2.4 – Electronic Mail in the Internet

❖ Components of an E-mail System

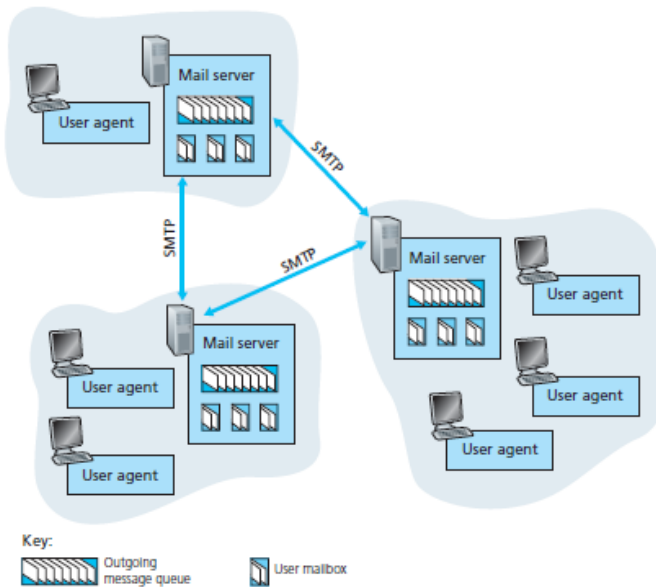


Figure 2.16 ♦ A high-level view of the Internet e-mail system

❖ How an E-mail system works? (Working Steps, Protocols)

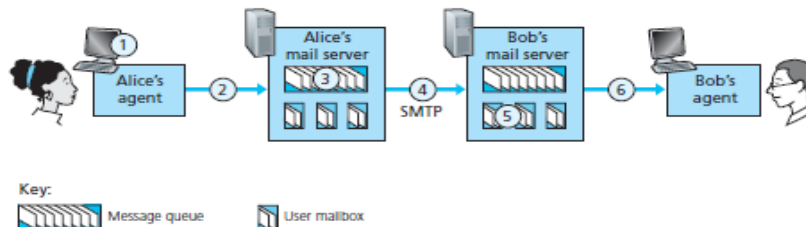


Figure 2.17 ♦ Alice sends a message to Bob

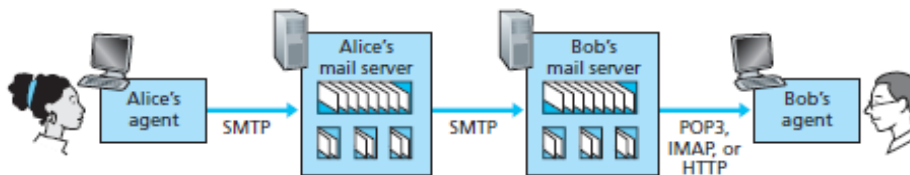
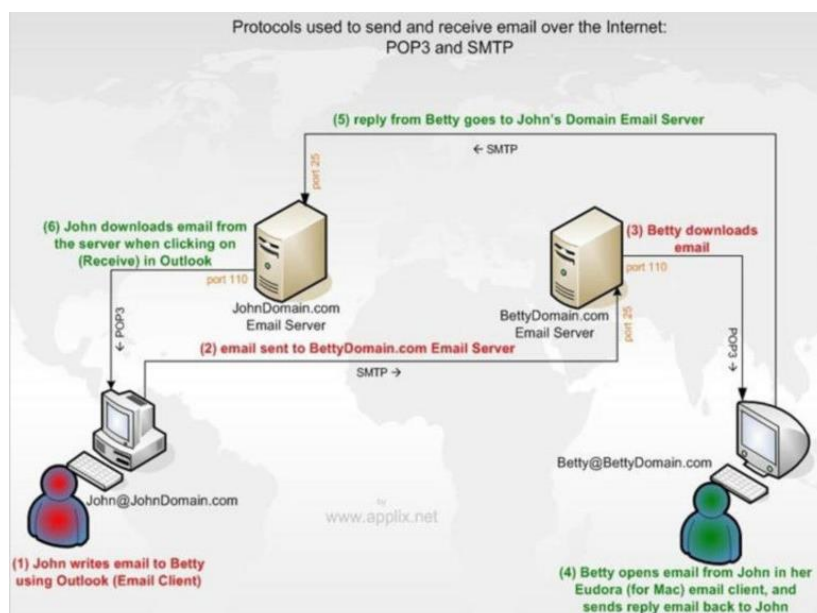
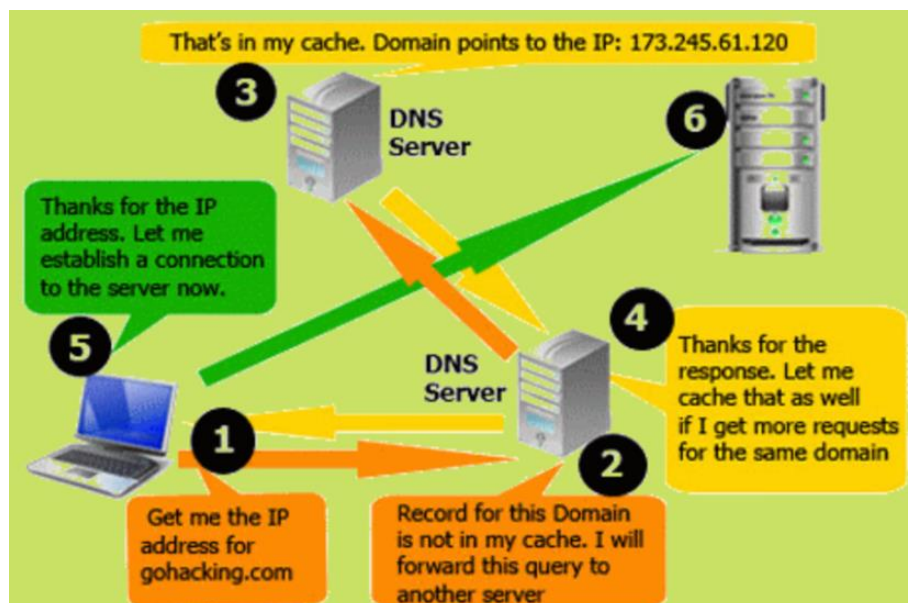
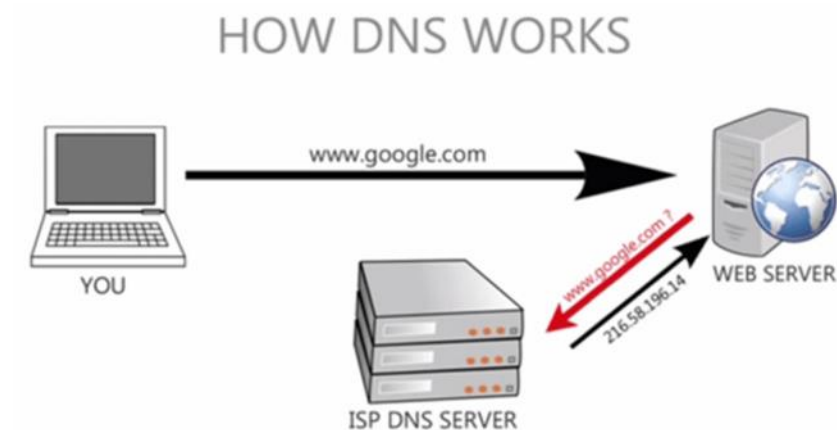


Figure 2.18 ♦ E-mail protocols and their communicating entities



Section 2.5 – Domain Name System (DNS)

- ❖ Services provided by DNS
- ❖ How DNS works?



- ❖ DNS hierarchy

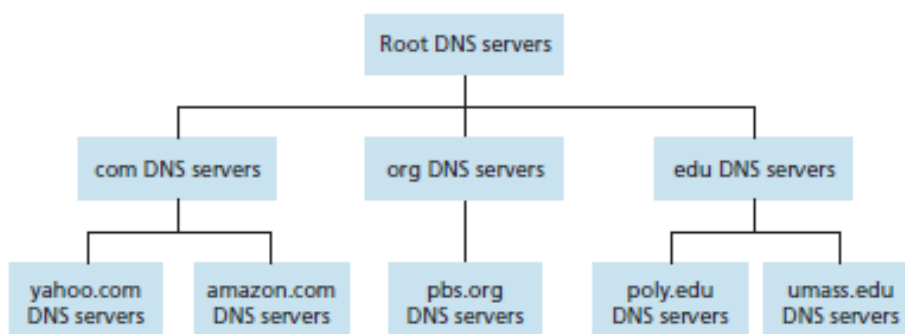
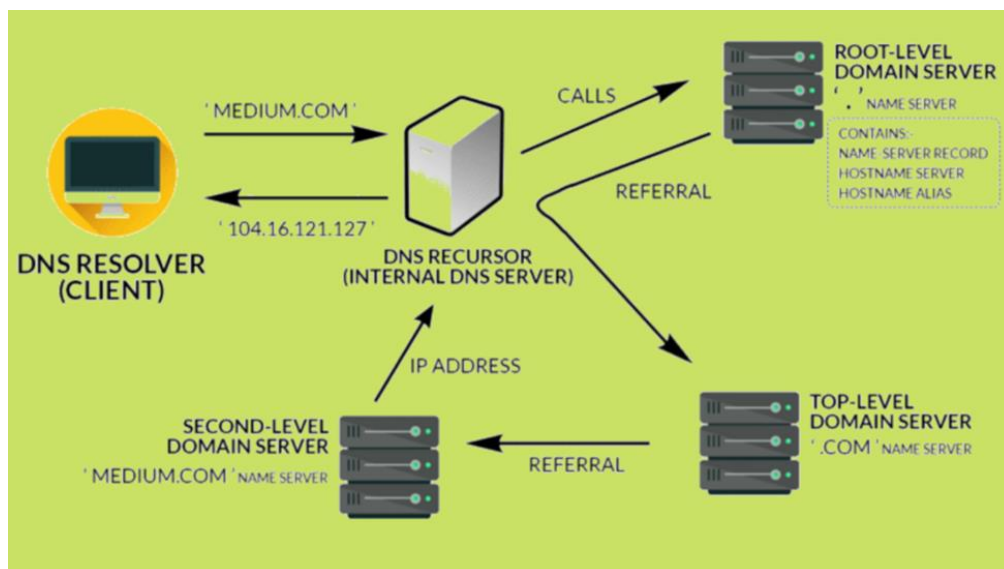


Figure 2.19 ♦ Portion of the hierarchy of DNS servers



- ✓ **Exercise 6:** Draw a **diagram** (containing different types/levels of DNS servers) to show the **steps of an iterated query** for a DNS request (Suppose there is no DNS cache) to get the name-to-IP-address mapping from a client PC connected to Internet.

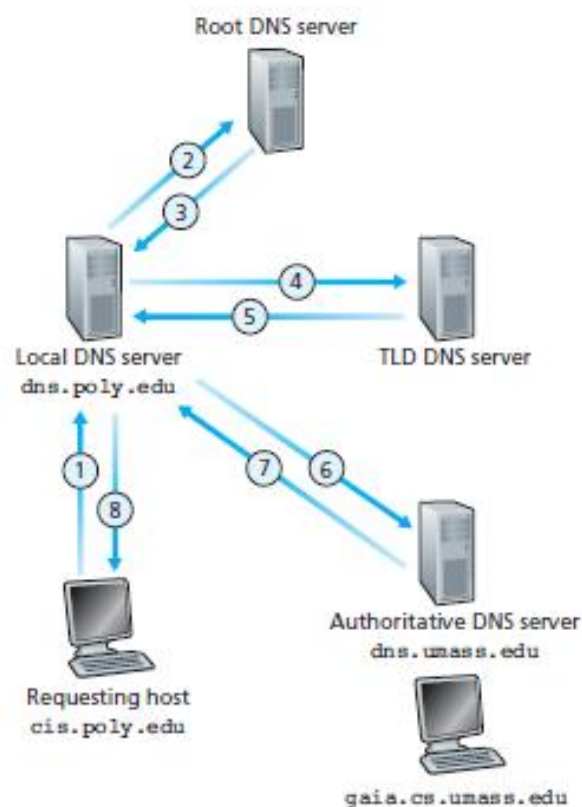


Figure 2.21 ♦ Interaction of the various DNS servers

❖ DNS records

DNS: distributed db storing resource records (RR)

RR format: (name, value, type, ttl)

type=A

- **name** is hostname
- **value** is IP address

type=NS

- **name** is domain (e.g., foo.com)
- **value** is hostname of authoritative name server for this domain

type=CNAME

- **name** is alias name for some “canonical” (the real) name
- **www.ibm.com** is really **servereast.backup2.ibm.com**
- **value** is canonical name

type=MX

- **value** is name of mailserver associated with **name**

✓ **Exercise 7:** Suppose you have a **Host C**, a **local name server L**, and **authoritative name servers** Aroot, Acom, and Agoogle.com, where the naming convention **Ax** means that the **name server knows about the name zone x**. Ax is a variable and NOT a hostname. Aroot is a root name server known to L, with IP address **198.41.0.4**. Assume that all name servers initially have nothing in their caches.

a) Using the **resource records** below, provide the hostnames and IP addresses for Acom and Agoogle.com

b) List the **sequence of DNS queries and corresponding resource records** that are exchanged when C wants to lookup the address for www.google.com. When a name server Ax contacts Ay, please specify the IP address for Ay. The following represents the level of detail we are looking for:

i) Ax contacts Ay at 1.2.3.4 to resolve www.foobar.com

ii) Ay returns resource records R1, R2, ..., Rn.

c) Client C performs a lookup for mail.google.com immediately after the previous request. Assume all records have long TTLs. List the sequence of DNS queries and corresponding resource records exchanged during the lookup.

Name Server Variable	Resource Record
A _{root}	{com, a.gtld-servers.net, NS, IN}
A _{root}	{a.gtld-servers.net, 192.5.6.30, A, IN}
A _{com}	{google.com, ns1.google.com, NS, IN}
A _{com}	{ns1.google.com, 216.239.32.10, A, IN}
A _{google.com}	{www.google.com, 66.102.7.104, A, IN}
A _{google.com}	{mail.google.com, 66.102.7.83, A, IN}

Knowledge Test # 3

Q.1 What are the **3 (Three)** main components of an **E-mail system**? Describe the steps how an e-mail system works.

Q.2 What are the protocols used as **mail forwarding protocol** and **mail access protocols**?

Q.3 Why **Domain Name System (DNS)** is used? What are the **different types of DNS servers**? What are the **problems** of using a **centralized DNS server** in Internet?

Q.4 Suppose you open a startup company "**flashBD**" and want to set up your company network. Your network has the **following servers**:

- **DNS server:** "**dns1.flashBD.com**" with IP as "128.119.1.40"
- **Web server:** "**flashBD.com**" with two IP as "128.119.1.55" and "128.119.1.56". Internet users can also access the web server by "**www.flashBD.com**".
- **Email server:** "**mailserver.flashBD.com**" with IP as "128.119.1.60"

Your company's email address is "**username@flashBD.com**".

a) What **resource records (RRs)** do you need to provide to the upper-level "**.com**" Registrar?

b) What **RRs** do you need to put in your **company's DNS server**?